ENEE 132 — Basic Electronics I (Semiconductor Devices)

Lecture 4 - Rectifiers

7 July 2026

Today's Agenda — Lecture Notes pp. 47–65

- **Part 1 — The DC Power Supply & Half-Wave Rectification**
  - The power-supply block diagram and how a half-wave rectifier works
  - Average value, the barrier-potential effect, PIV, and transformer coupling
- **Part 2 — Center-Tapped Full-Wave Rectifiers**
  - From half-wave to full-wave, and the center-tapped design
  - Output voltage, turns ratio, and PIV
- **Part 3 — The Bridge Full-Wave Rectifier**
  - Four-diode operation, output voltage, and PIV
  - Center-tapped vs. bridge, side by side
- **Part 4 — Wrap-Up**
  - Key takeaways and a look ahead to filters and regulators

Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**
  - Describe the blocks of a basic dc power supply and what each one does
  - Explain half-wave rectification and find the average (DC) value of the output
  - Account for the barrier potential and determine the peak output voltage
  - Define peak inverse voltage (PIV) and use it to choose a diode rating
  - Explain the center-tapped and bridge full-wave rectifiers and analyze their output and PIV
  - Compare the center-tapped and bridge designs and choose between them

Where We Left Off — Why Rectifiers?

- **Recap**
  - A diode conducts in one direction and blocks the other
  - That one-way behaviour is exactly what we need to convert AC into DC

- **The Big Picture**
  - Wall power is 120 V, 60 Hz AC (230 V, 50 Hz in Ghana); almost every electronic circuit needs steady DC
  - A rectifier is the stage that turns AC into pulsating DC
  - It is found in every DC power supply — from a phone charger to lab equipment

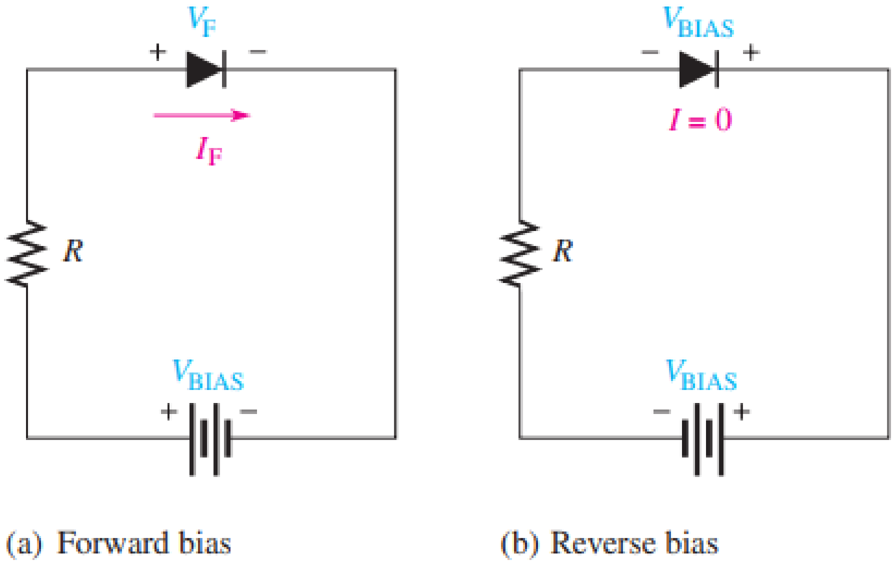


Fig 2.14 — Forward-bias and reverse-bias connections (diode symbol)

Part 1 — The DC Power Supply & Half-Wave Rectification

The Basic DC Power Supply

- **From the Wall to Steady DC**
  - A DC power supply converts alternating high voltage AC into a constant low voltage DC voltage
- **The Four Stages**
  - Transformer — steps the AC voltage up or down (and isolates the circuit)
  - Rectifier — converts AC to pulsating DC (half-wave or full-wave)
  - Filter — smooths the pulsations into nearly constant DC
  - Regulator — holds the DC steady despite changes in line voltage or load
  - The load is the circuit powered by the supply

The Basic DC Power Supply – Visualized

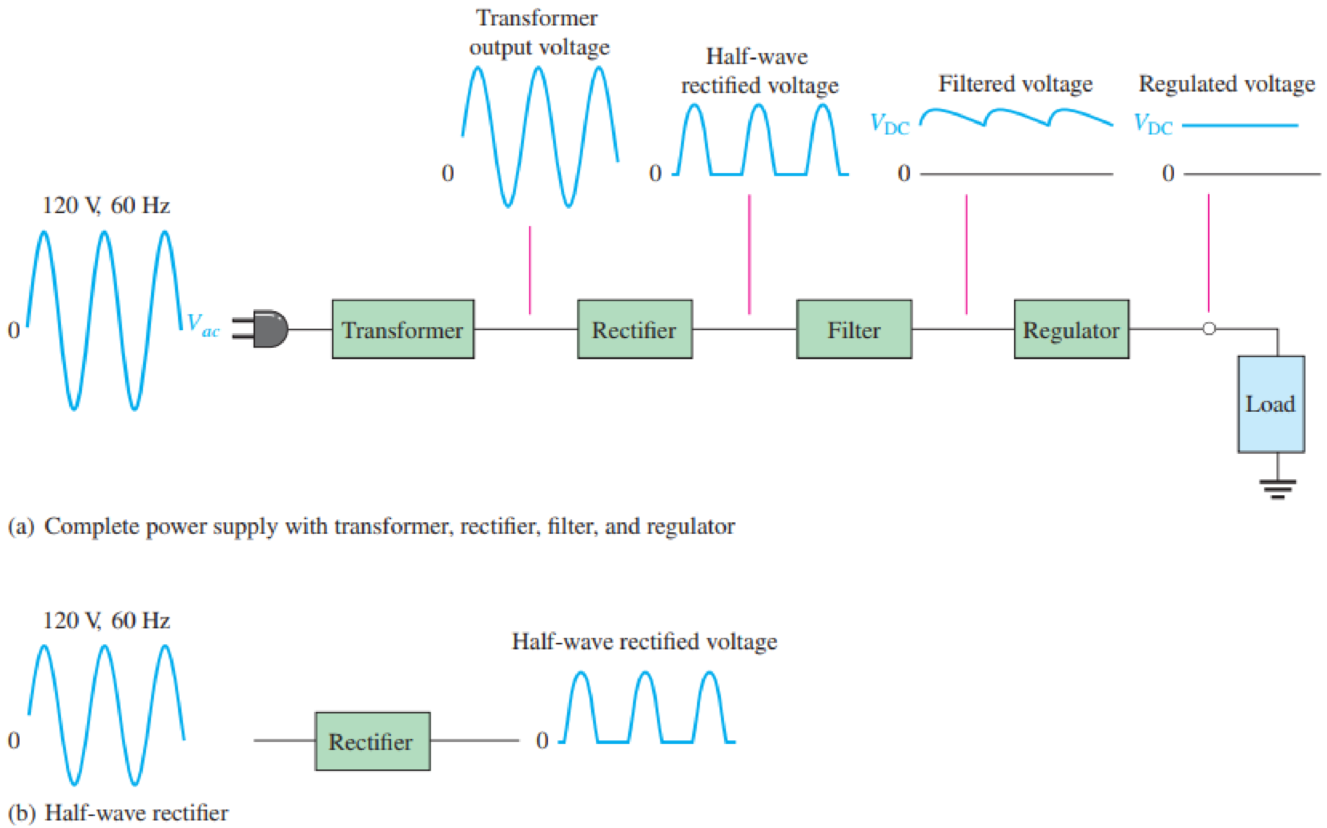
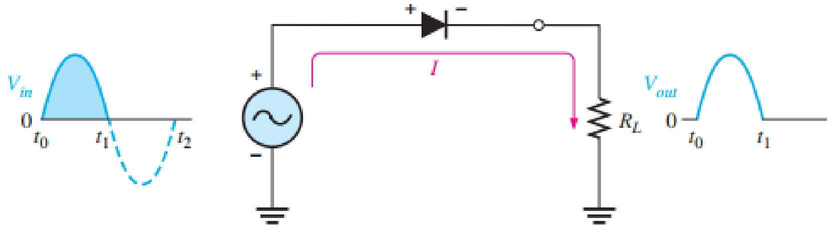


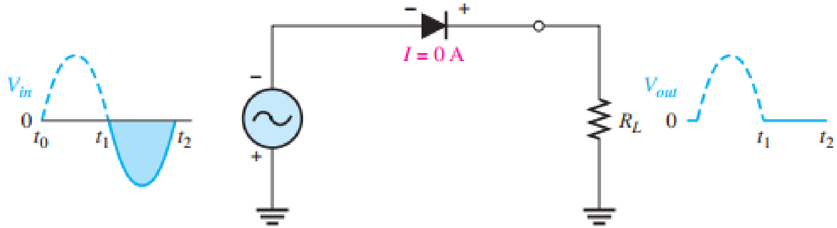
Fig 2.19 — Block diagram of a DC power supply: transformer, rectifier, filter, regulator, load.

Half-Wave Rectifier Operation

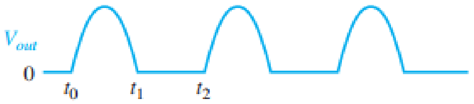
- **The Circuit**
  - One diode in series with a load resistor RL, driven by an AC source
- **One Cycle (Ideal Diode)**
  - Positive half-cycle: the diode is forward-biased and conducts → output follows the input across the load R_L
  - Negative half-cycle: the diode is reverse-biased → no current → output is 0 V
- **The Result**
  - Only the positive half-cycles reach the load — a pulsating DC voltage at 60 Hz



(a) During the positive alternation of the 60 Hz input voltage, the output voltage looks like the positive half of the input voltage. The current path is through ground back to the source.



(b) During the negative alternation of the input voltage, the current is 0, so the output voltage is also 0.



(c) 60 Hz half-wave output voltage for three input cycles

Fig 2.20 — Half-wave rectifier operation using the ideal diode model.

Average Value of the Half-Wave Output

- **What a DC Voltmeter Reads**
  - The average value is the area under the curve over one full cycle, divided by 2π
- **The Result**

$$V_{AVG} = \frac{V_p}{\pi}$$

  - So the DC value of a half-wave signal is only about 31.8% of its peak

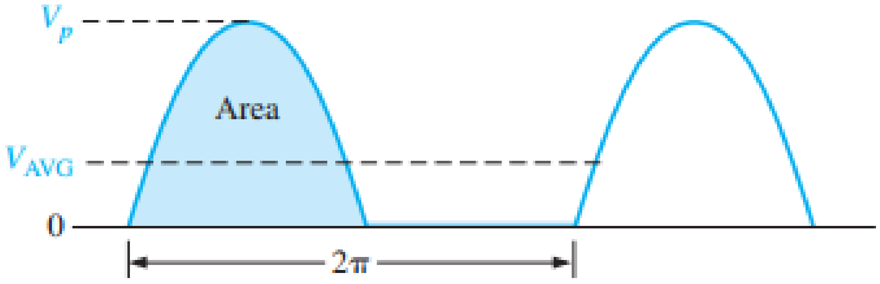


Fig 2.21 — Average value of the half-wave rectified signal.

Worked Example — Half-Wave Average Value

- **The Question (Example 2-2)**
  - A half-wave rectified voltage has peak value Vp. Find its average (DC) value.

Fig 2.22 — Half-wave rectified waveform for Example 2–2.

- **The Solution**

$$V_{AVG} = \frac{V_p}{\pi} = \frac{50\text{ V}}{\pi} = \mathbf{15.9\ V}$$

- **The Point**
  - A DC voltmeter reads only about a third of the peak (31.8% of Peak voltage)

Effect of the Barrier Potential on the Half-Wave Rectifier Output

- **Using the Practical Model**
  - The input must first overcome the diode's 0.7 V barrier before it conducts
  - So the output peak is 0.7 V below the input peak: Vp(out) = Vp(in) – 0.7 V
- **Ideal vs. Practical**
  - When the peak is much larger than 0.7 V (≥ 10 V rule of thumb), the ideal model is fine
  - Example: a 5 V peak → 4.3 V (the 0.7 V matters); a 100 V peak → 99.3 V (barely matters)

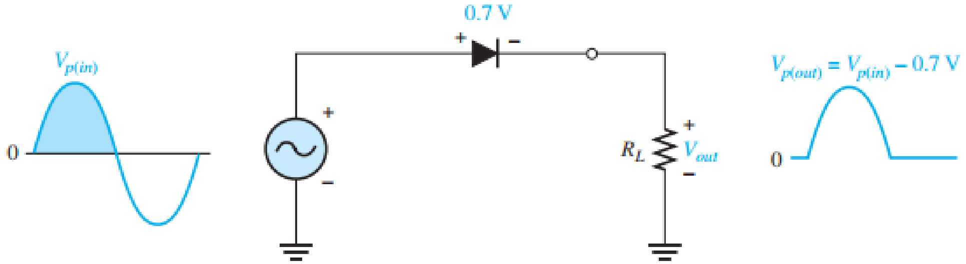


Fig 2.23 — The barrier potential reduces the peak output by about 0.7 V.

Worked Example — The 0.7 V Drop in Two Cases

- **Example 2–3 — Two Rectifiers, the Same 0.7 V Drop**
  - Circuit (a): 5 V peak input with a 1N4001 diode
  - Circuit (b): 100 V peak input with a 1N4003 diode
  - Draw the output voltages of each rectifier for the indicated input voltages

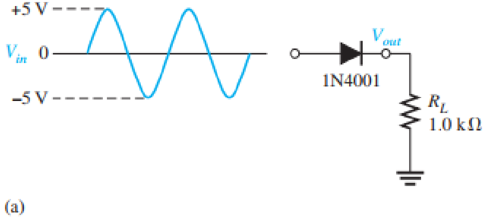


Fig 2.24 — Circuits for Example 2–3 (1N4001 and 1N4003).

- **Solution**
  - (a) Vp(out) = 5 – 0.7 = 4.30 V
  - (b) Vp(out) = 100 – 0.7 = 99.3 V

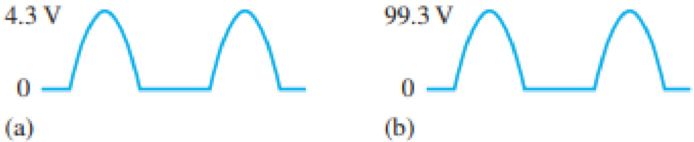


Fig 2.25 — Output voltages for the circuits

Peak Inverse Voltage (PIV)

- **The Reverse Stress**
  - On each negative half-cycle the diode is reverse-biased and must withstand the input peak
  - This maximum repetitive reverse voltage is the peak inverse voltage: PIV = Vp(in)
- **Choosing a Diode**
  - The diode's reverse-voltage rating must exceed the PIV
  - Rule of thumb: the diode should be rated at least 20% higher than the PIV

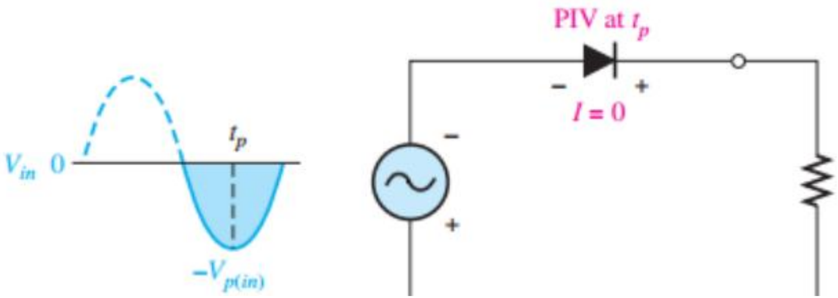


Fig 2.26 — PIV occurs at the negative peak, when the diode is reverse-biased.

Transformer Coupling

- **Why Use a Transformer**
  - Steps the AC voltage up or down as needed
  - Electrically isolates the rectifier from the wall supply (a safety advantage)
- **Turns Ratio**
  - Turns ratio n = (secondary turns) / (primary turns)
  - The secondary voltage of a transformer equals the turns ratio, *n*, times the primary voltage.
  - Vsec = *n* · Vpri — so n > 1 it is step-up transformer, if *n* < 1 it is a step-down transformer. If *n* = 1, it is an isolation transformer
- **Rectifier Output**
  - Vp(out) = Vp(sec) – 0.7 V
- **PIV**
  - PIV = Vp(sec)

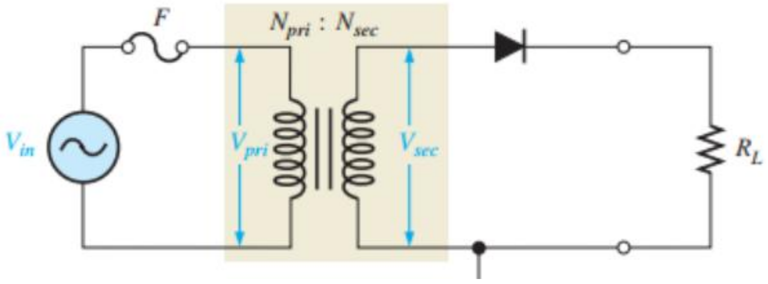


Fig 2.27 — Half-wave rectifier with transformer-coupled input voltage.

Worked Example — Transformer-Coupled Half-Wave

- **The Question (Example 2-4)**
  - Determine the peak value of the output voltage for Figure 2–28 if the turns ratio is 0.5.
- **The Solution**
  - Peak secondary voltage: $V_{p(sec)} = n \cdot V_{p(pri)} = 0.5 \times 170\text{ V} = 85\text{ V}$
  - Peak output (one diode drop): $V_{p(out)} = V_{p(sec)} - 0.7\text{ V} = 85 - 0.7 = 84.3\text{ V}$
- **Note**
  - $n < 1$ steps the voltage down; the diode's PIV equals $V_{p(sec)} = 85\text{ V}$
- **Related Problem**
  - Determine the peak value of the output voltage for Figure 2–28 if $n = 2$ and $V_{p(in)} = 312$ V.
  - What is the PIV across the diode?
  - Describe the output voltage if the diode is turned around.

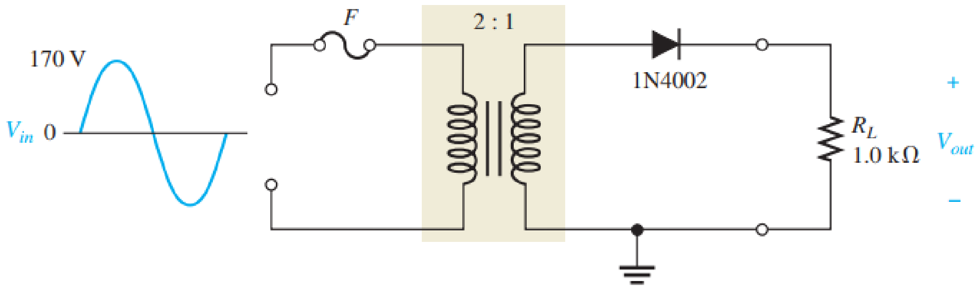


Fig 2.28 — Circuit for Example 2–4.

Quick Check #1 — Half-Wave Rectification

- **Question 1**
  - Name the four stages of a basic dc power supply, in order.
- **Question 2**
  - During which half-cycle does a positive half-wave rectifier conduct?
- **Question 3**
  - What is the average (DC) value of a half-wave output in terms of Vp?
- **Question 4**
  - How does the barrier potential change the peak output voltage?
- **Question 5**
  - What is PIV, and how does it relate to the input peak?

Part 2 — Full-Wave Rectifiers

From Half-Wave to Full-Wave

- **The Idea**
  - A half-wave rectifier wastes one half of every cycle
  - A full-wave rectifier uses BOTH half-cycles — current flows through the load on each one

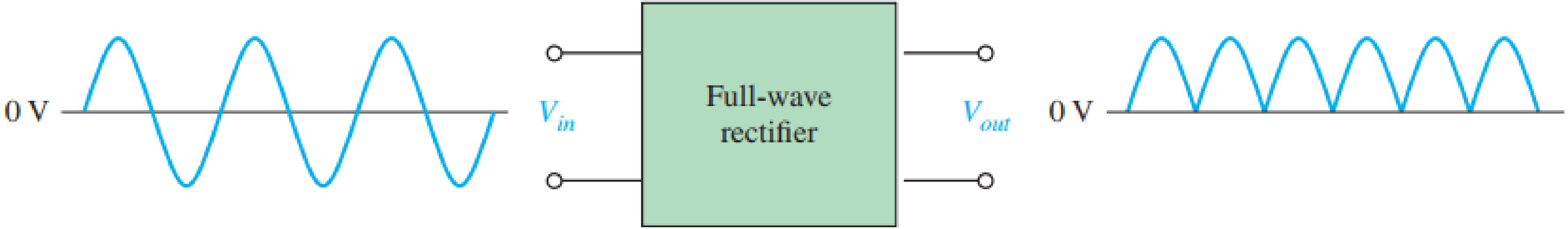


Fig 2.29 — Full-wave rectification: output frequency is twice the input.

- **The Result**
  - The average (DC) value doubles:

$$V_{AVG} = \frac{2V_p}{\pi}$$

  - That's about 63.7% of the peak — twice the half-wave value

Worked Example — Full-Wave Average Value

- **The Question (Example 2-5)**
  - Find the average (DC) value of a full-wave rectified voltage.

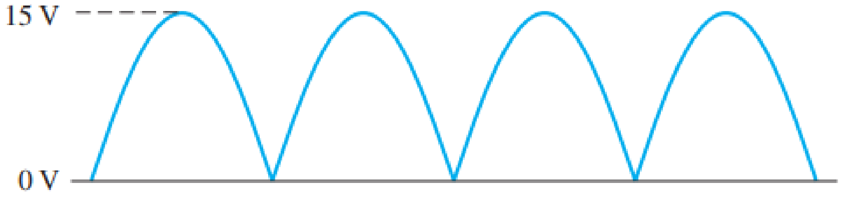


Fig 2.30 — Full-wave rectified waveform for Example 2–5.

- **The Solution**

$$V_{AVG} = \frac{2V_p}{\pi} = \frac{2(15\text{ V})}{\pi} = \mathbf{9.55\ V}$$

- **Compare**
  - That's about 63.7% of the peak

The Center-Tapped Full-Wave Rectifier

- **The Circuit**
  - A center-tapped rectifier is a type of full-wave rectifier that uses two diodes connected to the secondary of a center-tapped transformer
  - Two diodes fed from a center-tapped transformer secondary
  - Half of the total secondary voltage appears between the center tap and each end
- **How It Works**
  - Positive half-cycle: D1 forward-biased, D2 reverse-biased → current through D1 and RL
  - Negative half-cycle: D2 forward-biased, D1 reverse-biased → current through D2 and RL
  - Current flows the same direction through RL on both halves → full-wave output

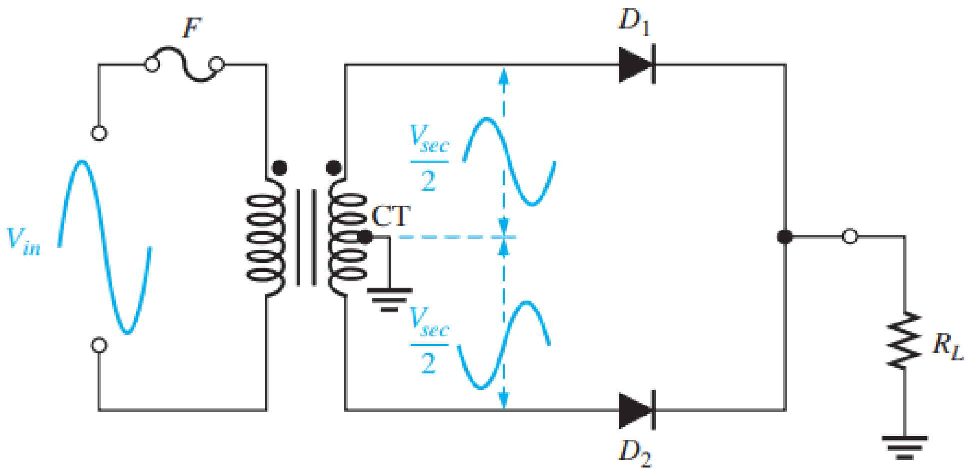
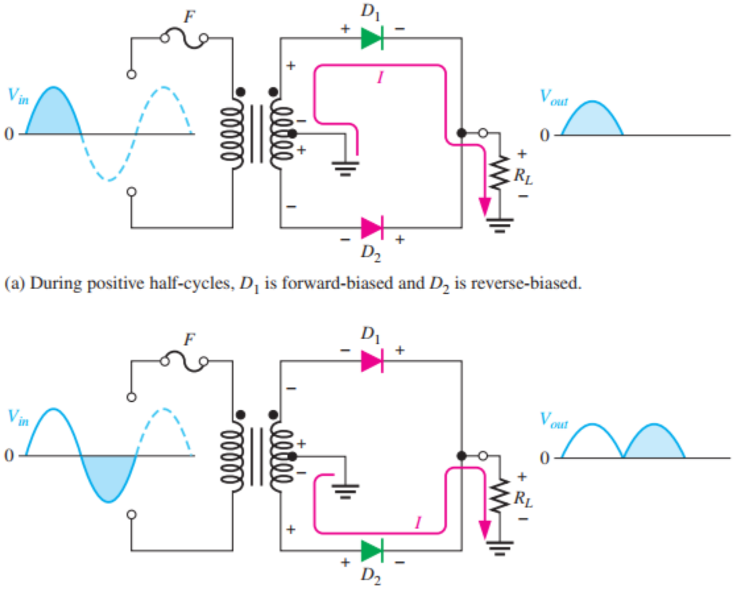


Fig 2.31 — A center-tapped full-wave rectifier.

Center-Tapped — How It Works

- **Two Diodes Share the Work — One per Half-Cycle**
  - Positive half-cycle: D1 forward-biased, D2 reverse-biased → current through D1 and RL
  - Negative half-cycle: D2 forward-biased, D1 reverse-biased → current through D2 and RL
  - In BOTH half-cycles the load current is in the same direction → a full-wave output
  - Each half-cycle delivers a pulse, so the output frequency is twice the input



(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.

(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

Fig 2.32 — Operation of a center-tapped rectifier over both half-cycles.

Center-Tapped — Effect of Turns Ratio on the Output Voltage

- **With n = 1 (an Isolation Transformer)**
  - The total secondary voltage equals the primary: Vp(sec) = Vp(pri)
  - Only HALF the secondary drives each diode path: Vp(sec)/2
  - So the peak output is: Vp(out) = Vp(pri)/2 – 0.7 V
- **Why This Matters**
  - A turns ratio of 1 gives an output near half the input — to recover the full input peak you step up (next slide, n = 2)

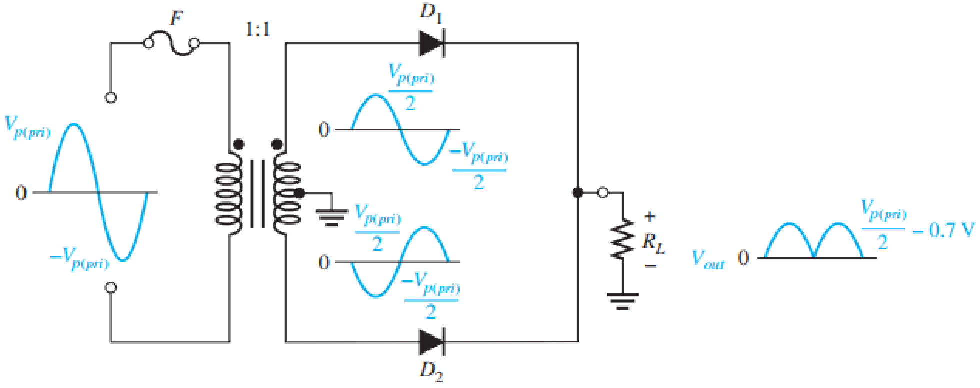


Fig 2.33 — Center-tapped rectifier with a transformer turns ratio of 1.

Center-Tapped: Output Voltage & Turns Ratio

- **The Output Rule**
  - The output is always half the total secondary voltage, less one diode drop:

$$V_{out} = \frac{V_{sec}}{2} - 0.7\ V$$

- **Turns-Ratio Examples**
  - n = 1: each half of the secondary gets half the primary peak
  - n = 2 (step-up): the total secondary is twice the primary, so each half equals the primary peak

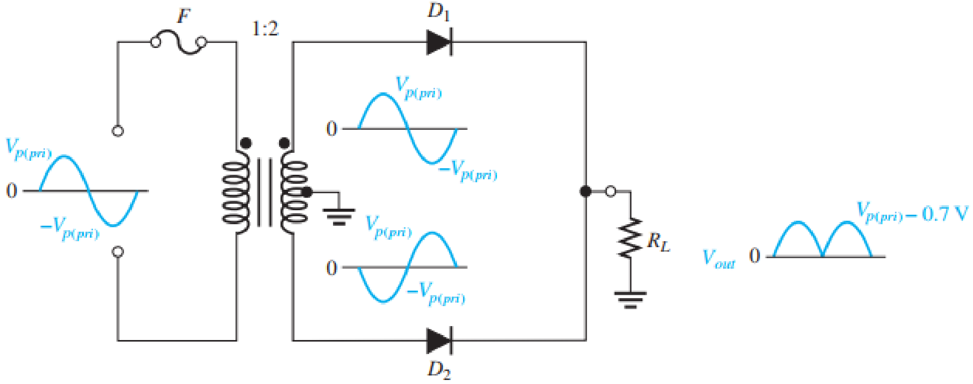


Fig 2.34 — Center-tapped rectifier with a transformer turns ratio of 2.

Center-Tapped: Peak Inverse Voltage

- **The Reverse Stress**
  - Each diode alternates between forward- and reverse-biased
  - The reverse-biased diode must withstand nearly the full secondary voltage
- **The Formula**
  - The peak inverse voltage across D2 is

$$\text{PIV} = \left(\frac{V_{p(sec)}}{2} - 0.7\text{ V}\right) - \left(-\frac{V_{p(sec)}}{2}\right) = \frac{V_{p(sec)}}{2} + \frac{V_{p(sec)}}{2} - 0.7\text{ V}$$
$$= V_{p(sec)} - 0.7\text{ V}$$

  - The PIV across either diode in a full-wave center-tapped rectifier is

PIV = 2Vp(out) + 0.7V

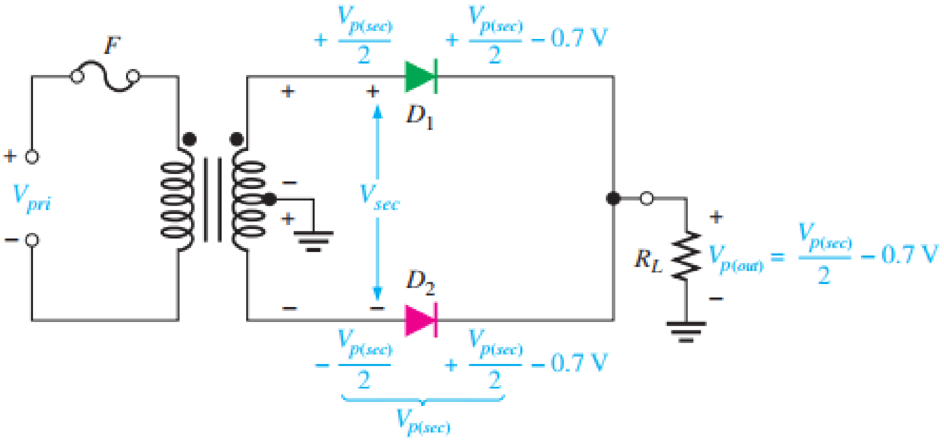


Fig 2.35 — Diode reverse voltage (D2 reverse-biased, D1 forward-biased).

Worked Example — Center-Tapped Rectifier

- **The Question (Example 2-6)**
  - (a) Show the voltage waveforms across each half of the secondary winding and across R_L when a 100 V peak sine wave is applied to the primary winding in Figure 2–36.
  - (b) Assuming a 20% margin, what minimum PIV rating must the diodes have?

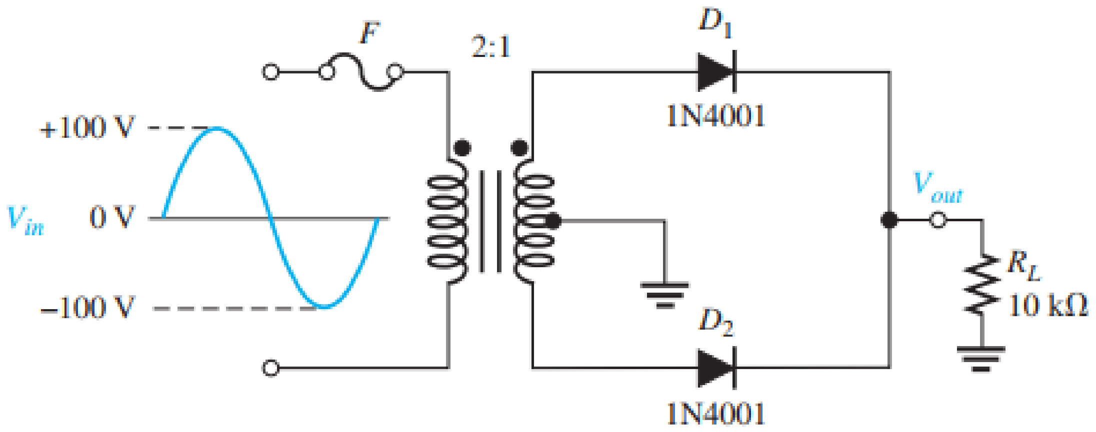


Fig 2.36 — Circuit for Example 2–6.

Worked Example — Solution

- **(a) Example 2–6 — 100 V Peak Input, n = 0.5**
  - Total peak secondary: $V_{p(sec)} = 0.5 \times 100 = 50$ V
  - Across each half of the secondary: $50 / 2 = 25$ V peak
  - Peak load output: $V_{p(out)} = 25 - 0.7 = 24.3$ V

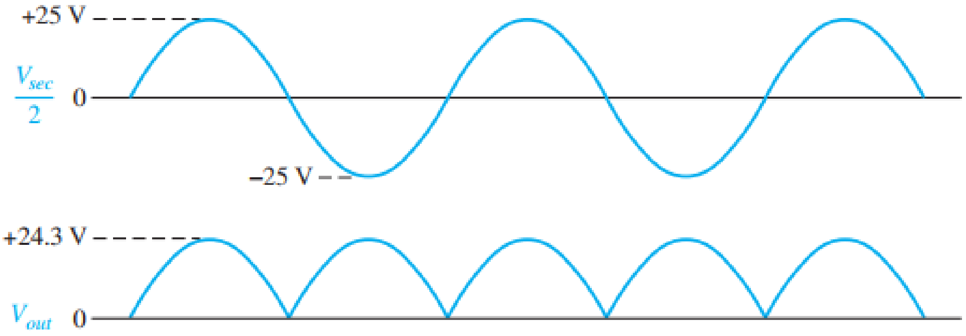


Fig 2.37 — Output waveforms for Example 2–6.

- **(b) Required Diode PIV**
  - $PIV = 2 \cdot V_{p(out)} + 0.7 = 2(24.3) + 0.7 = 49.3$ V
  - With a 20% margin, choose diodes rated at least 60 V

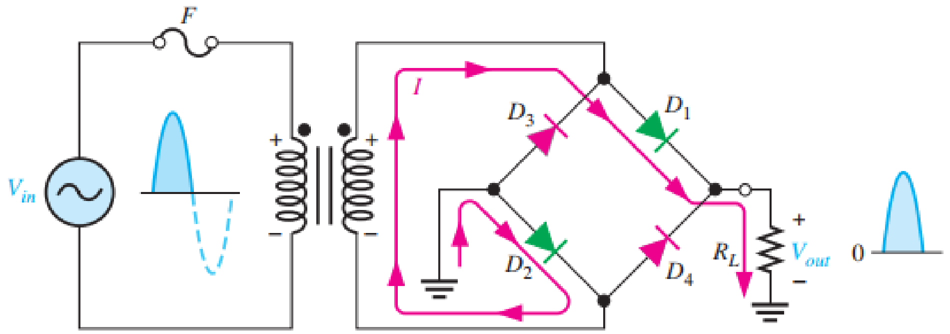
Quick Check #2 — Full-Wave & Center-Tapped

- **Question 1**
  - What is the average (dc) value of a full-wave output in terms of Vp?
- **Question 2**
  - In a center-tapped rectifier, which diode conducts on the positive half-cycle?
- **Question 3**
  - Write the output-voltage formula for a center-tapped rectifier.
- **Question 4**
  - Write the PIV formula for a center-tapped rectifier.
- **Question 5**
  - Why does a full-wave output have twice the dc value of a half-wave output?

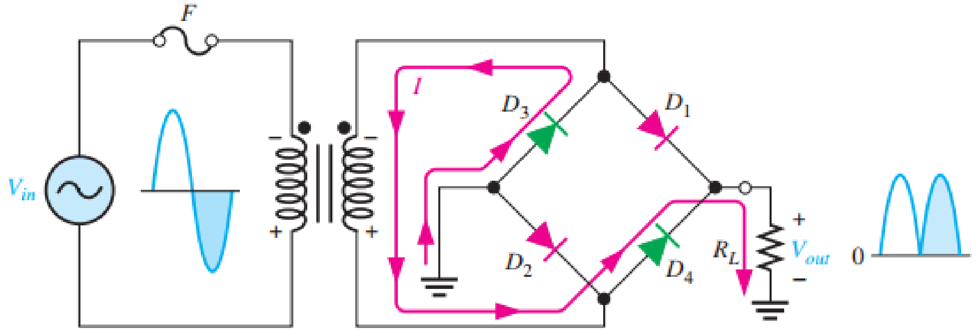
Part 3 — The Bridge Rectifier

The Bridge Full-Wave Rectifier

- **The Circuit**
  - The bridge rectifier uses four diodes arranged in a bridge — no center-tapped transformer needed
- **How It Works**
  - Positive half-cycle: D1 and D2 are forward biased and conduct (D3, D4 off) → output across RL
  - Negative half-cycle: D3 and D4 are forward biased and conduct (D1, D2 off) → current the same direction through RL
  - Both halves drive the load the same way → full-wave output



(a) During the positive half-cycle of the input, D_1 and D_2 are forward-biased and conduct current. D_3 and D_4 are reverse-biased.



(b) During the negative half-cycle of the input, D_3 and D_4 are forward-biased and conduct current. D_1 and D_2 are reverse-biased.

Fig 2.38 — Operation of a bridge rectifier over both half-cycles.

Bridge: Output Voltage

- **Two Diodes in the Path**
  - On each half-cycle, two diodes are in series with the load
  - So two diode drops (2 × 0.7 V) come off the secondary peak:

    $V_{p(out)} = V_{p(sec)} - 1.4\ V$
- **Compared to Center-Tapped**
  - The bridge uses the FULL secondary voltage (not half), so no step-up transformer is needed for the same output

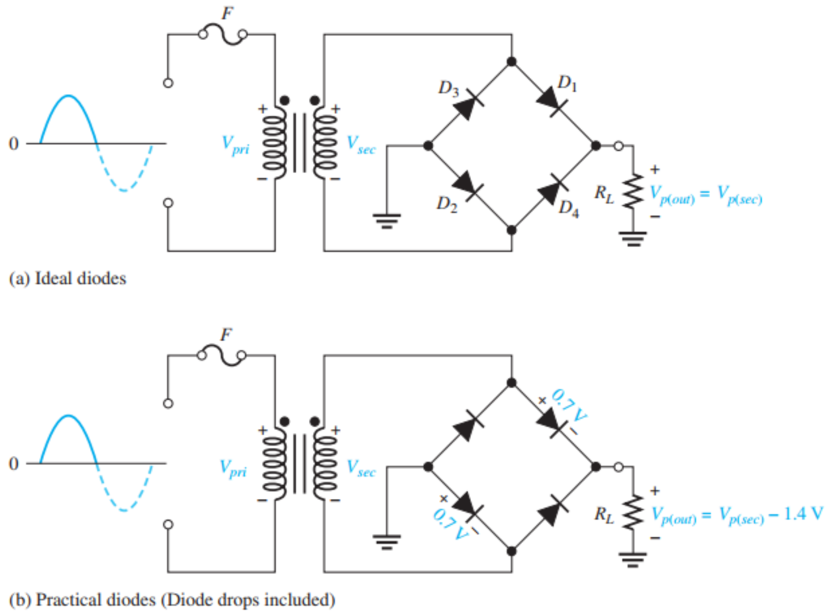
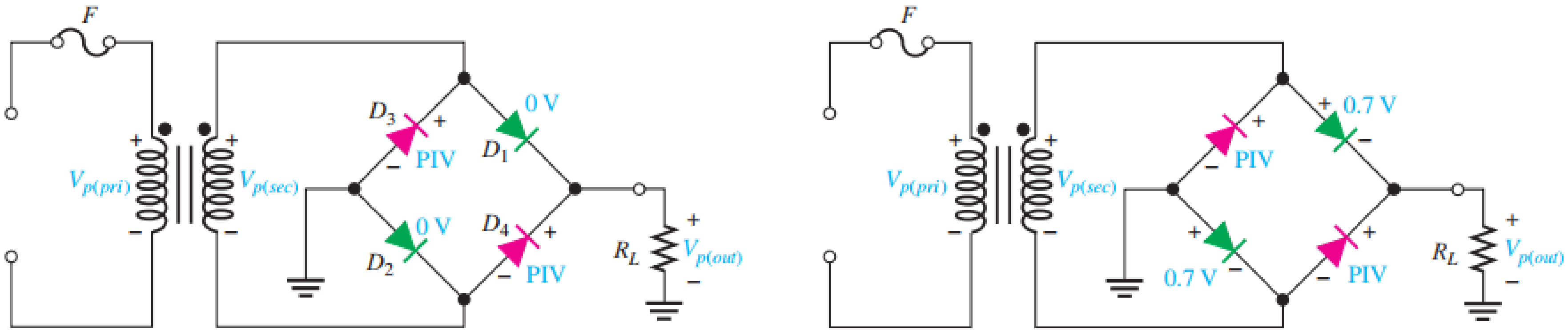


Fig 2.39 — Bridge operation during a positive half-cycle.

Bridge: Peak Inverse Voltage

- **The Reverse Stress**
  - Each reverse-biased diode sees the peak secondary voltage
  - In terms of the output: PIV = Vp(out) + 0.7 V



(a) For the ideal diode model (forward-biased diodes D_1 and D_2 are shown in green), PIV = $V_{p(out)}$.

(b) For the practical diode model (forward-biased diodes D_1 and D_2 are shown in green), PIV = $V_{p(out)}$ + 0.7 V.

Fig 2.40 — Peak inverse voltage across D3 and D4 in a bridge rectifier.

- **The Big Advantage**
  - That's about HALF the PIV of a center-tapped rectifier for the same output voltage
  - So bridge diodes can have a lower voltage rating

Worked Example — Bridge Rectifier

- **The Question (Example 2-7)**
  - Determine the peak output voltage for the bridge rectifier in Figure 2–41. Assuming the practical model, what PIV rating is required for the diodes? The transformer is specified to have a 12 V rms secondary voltage for the standard 120 V across the primary.
- **The Solution**
  - Secondary peak: Vp(sec) = 1.414 × 12 V = 17 V
  - Output peak: Vp(out) = Vp(sec) − 1.4 V = 17 − 1.4 = 15.6 V
  - PIV = Vp(out) + 0.7 V = 15.6 + 0.7 = 16.3 V → the diode rating must exceed this

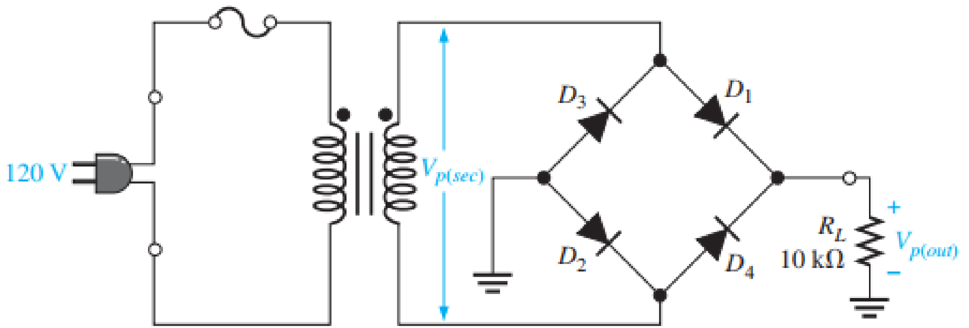


Fig 2.41 — Circuit for Example 2–7 (12 V rms secondary).

Center-Tapped vs. Bridge — Side by Side

- **Center-Tapped**
  - Two diodes + a center-tapped transformer
  - Output: $V_{p(out)} = V_{p(sec)}/2 - 0.7\ V$ (uses half the secondary, one diode drop)
  - $PIV = 2V_{p(out)} + 0.7\ V$ — higher reverse stress on each diode
- **Bridge**
  - Four diodes, an ordinary (non-tapped) transformer
  - Output: $V_{p(out)} = V_{p(sec)} - 1.4\ V$ (uses the full secondary, two diode drops)
  - $PIV = V_{p(out)} + 0.7\ V$ — about half the center-tapped value
- **The Trade-Off**
  - Bridge: two more diodes and an extra diode drop, but lower PIV and no center tap — the most common choice

Quick Check #3 — Bridge & Comparison

- **Question 1**
  - How many diodes does a bridge rectifier use, and how many conduct at once?
- **Question 2**
  - Write the bridge output-voltage formula (practical model).
- **Question 3**
  - Write the bridge PIV formula.
- **Question 4**
  - For the same output, how does the bridge PIV compare with the center-tapped PIV?
- **Question 5**
  - Name one advantage and one disadvantage of the bridge versus the center-tapped design.

Part 4 — Wrap-Up

Key Takeaways from Today

- A dc power supply has four stages: transformer → rectifier → filter → regulator → (load)
- Half-wave: uses one half-cycle; $V_{AVG} = V_p/\pi \approx 0.318\ V_p$; pulses at 60 Hz
- Practical model: $V_{p(out)} = V_{p(in)} - 0.7$ V; $PIV = V_{p(in)}$, and rate the diode ~20% higher
- Full-wave: uses both half-cycles; $V_{AVG} = 2V_p/\pi \approx 0.637\ V_p$; pulses at 120 Hz
- Center-tapped: 2 diodes; $V_{p(out)} = V_{p(sec)}/2 - 0.7$ V; $PIV = 2V_{p(out)} + 0.7$ V
- Bridge: 4 diodes; $V_{p(out)} = V_{p(sec)} - 1.4$ V; $PIV = V_{p(out)} + 0.7$ V — about half the center-tapped PIV
- Transformer coupling sets the voltage ($V_{sec} = n \cdot V_{pri}$) and provides isolation

Coming Up — Week 5: Filters & Voltage Regulators

- **Next Topic — Power-Supply Filters & Regulators (Lecture Notes pp. 66–74)**
  - Smoothing pulsating dc with a capacitor-input filter
  - Ripple voltage and the ripple factor
  - Voltage regulation against line and load changes
- **To Prepare**
  - Be comfortable finding the peak output of a rectifier — we'll filter that output
  - Recall the difference between half-wave (60 Hz) and full-wave (120 Hz) ripple
- **Reminder**
  - Read pp. 66–74 before class

Thank You — Questions?